

**i Front page TIØ4145 2025**  
**Examination paper for TIØ4145 Corporate Finance**  
**Date: 11.12.2025**  
**Time: 15.00-19.00**

**Course contact: Maria Lavrutich**  
**Present at the exam location: No**

**Permitted examination support material:** D - No printed or hand-written support material is allowed. A specific basic calculator is allowed.

**OTHER INFORMATION**

**Read the questions carefully** and make your own assumptions. In your answers, explain clearly what assumptions you have made and how you have understood or limited the assignment.

If there are direct errors or omissions in the assignment set and you cannot make your own assumptions, please refer to the information about complaints regarding formal errors on the NTNU website "Explanation of grades and appeals".

**SPECIFIC INFORMATION FOR YOUR COURSE**

**Paper drawings:** All questions should be answered directly in Inspera. However, for sections 2,3, and 4 you may choose to answer on handwritten sheets. At the bottom of the question, you will find a seven-digit code. Fill in this code in the top left corner of the sheets you wish to submit.

**Please note that only blue or black ballpoint pens are allowed**

We recommend that you do this during the exam. If you require access to the codes after the examination time ends, click "Show submission".

You are responsible for filling in the correct codes on the handwritten sheets. Therefore, read the cover sheet carefully. The Examination Office cannot guarantee that incorrectly completed sheets will be added to your assignment.

**Weighting:** Multiple choice questions in Section 1 are graded as follows: 2 points are given for correct answer; 1 point is subtracted for wrong answer; 0 points is given when the question is left unanswered. Only one answer is correct.

**Withdrawing from the exam:**

If you wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click "Submit blank". This cannot be undone, even if the test is still open.

**Access to your answers:**

After the exam, you can find your answers under previous tests in Inspera. Be aware that it may take a working day until any hand-written material is available in "previous tests".

## 1.1 Beta

What does a *beta of a stock relative to the market* measure?

Select one alternative:

- ☐ Total risk
- ☐ Diversifiable risk
- ☐ Systematic risk
- ☐ Overall volatility of the financial markets



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Maximum marks: 2

## 1.2 Private Equity Funds

What differentiates private equity funds from other types of funds?

Select one alternative:

- ☐ They only rely on Leveraged Buyout (LBO) models as their valuation methodology
- ☐ They only use debt in their investments
- ☐ They invest in unlisted companies
- ☐ They primarily focus on short-term trading in public equity markets



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Maximum marks: 2

### 1.3 Risk-Neutral Probabilities

TunnelVision AS has 12 million shares outstanding with a current share price of \$40 per share. In one year, the share price is equally likely to be \$60 or \$30. The risk-free interest rate is 5%. What is the risk-neutral probability that TunnelVision AS's stock price will increase?

Select one alternative:

- ☐ 0.1
- ☐ 0.25
- ☒ 0.4
- ☐ 0.6



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Maximum marks: 2

### 1.4 Payout policy

What is a firm's payout policy?

Select one alternative:

- ☐ The rules governing how a firm must disclose information to shareholders.
- ☒ The way a firm chooses between alternative methods of distributing free cash flow to its equity holders.
- ☐ The decision of whether the firm should invest in new projects rather than keep cash internally.
- ☐ The method a firm uses to determine its capital structure.

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Maximum marks: 2

## 1.5 Solvency II

Which of the following statements about Solvency II is FALSE?

Select one alternative:

- ☐ Solvency II is the prudential regime for insurance and reinsurance undertakings in the EU.
- ☐ Solvency II is a regulatory framework that applies to all financial institutions in the EU, including banks, investment firms, and insurance companies. ✓
- ☐ Solvency II has a risk-based approach that enables to assess the “overall solvency” of insurance and reinsurance undertakings through quantitative and qualitative measures
- ☐ Solvency II sets out requirements to ensure the adequate protection of policyholders and beneficiaries.

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Maximum marks: 2

## 1.6 Option pricing

In most securities (e.g., stocks, corporate bonds), we estimate a required return (for example using CAPM) and discount expected cash flows at that rate. Why do we not typically price options by taking the expected payoff and discounting it at a CAPM-based discount rate?

Select one alternative:

- ☐ Because an option's payoff is a nonlinear function of the underlying asset, its risk and beta change across states and over time, so a single constant required return for the option is not consistent with its risk profile. ✓
- ☐ Because option payoffs depend on the underlying asset's price, any risk is already fully reflected in the stock's required return, so the option itself has no additional risk and should be discounted at the risk-free rate.
- ☐ Because option prices are very volatile, so it is impossible to compute any expected return for them.
- ☐ Because CAPM assumes a linear relation between returns and the market, it cannot be used for any derivative or contingent claim.

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Maximum marks: 2

## 1.7 CAPM

Which of the following statements is FALSE?

Select one alternative:

- ☐ Even naive investors with no information should hold the market portfolio.
- ☐ A conclusion of the CAPM is that investors should hold the market portfolio even if they do not have high trading skills.
- ☐ A conclusion of the CAPM is that investors should hold the market portfolio only if they have high quality information. ✓
- ☐ The CAPM assumption of homogeneous expectations is not necessarily a good description of the real world.

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Maximum marks: 2

## 1.8 Index Investing

Which statement best describes the key difference between mutual funds (including index funds) and exchange traded funds (ETFs)?

Select one alternative:

- ☐ Unlike mutual funds, an ETF trades like a common stock on a stock exchange. ✓
- ☐ Mutual funds and ETFs both trade continuously on stock exchanges, but ETFs charge higher fees because they track more complex indices.
- ☐ ETFs represent ownership in only one stock at a time, whereas mutual funds must hold a diversified portfolio.
- ☐ Mutual funds always track a benchmark index, while ETFs can only be actively managed.

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Maximum marks: 2

## 1.9 Share repurchases

Under what condition is a share repurchase a good use of cash for the shareholders who continue to hold their shares after the repurchase?

Select one alternative:

- ☐ It is always beneficial, since all shareholders gain equally from a repurchase, regardless of whether the stock is over- or undervalued.
- ☐ When managers want to signal confidence by distributing cash, even if they think the stock is overvalued.
- ☐ When management believes the stock is undervalued. ✓
- ☐ When the firm wants to increase the number of shares outstanding.

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Maximum marks: 2

## 1.10 Default and Bankruptcy in a Perfect Market

Which of the following statements is FALSE?

Select one alternative:

- ☐ An important consequence of leverage is the risk of bankruptcy.
- ☐ Whether default occurs depends on the cash flows, not on the relative values of the firm's assets and liabilities. ✓
- ☐ Economic distress is a significant decline in the value of a firm's assets, whether or not it experiences financial distress due to leverage.
- ☐ In a perfect capital market, Modigliani–Miller capital-structure irrelevance still holds even when debt is risky and the firm may default

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Maximum marks: 2